
IT206: Data Structure Lab using OOP (Winter '26)

Instructor: Dr. Sandip Modha, 2201, Faculty Block-2 [Sandip_modha@dau.ac.in]
Prerequisites: Introduction to Programming (C)
Course Placement: B.Tech. II Semester - Section A / B | Core
Course Credits: 2 Credits (1-0-2)
Associated Theory Course: IT205 / Data Structures 3 (3-0-0).
Foundation for: CS / IT stream of courses
Lectures (1x3) LT-02: TBD
TA contact info: TBD

Course Description:

This course aims to prepare students to implement different data structures that are taught in the IT205: Data Structures course, using the C++ programming language. The course begins with programming paradigms and gradually introduces essential object-oriented concepts such as classes, objects, encapsulation, constructors, destructors, inheritance, and polymorphism. These object-oriented concepts are essential to implement linear data structures such as stacks and queues and non-linear data structures including binary trees, heaps, and graphs.

Resource and Textbook:

- 'Data Structures and Algorithms in C++,' by Goodrich, Tamassia (Wiley, 2011).
- Other Materials or References will be provided in due course in the lab

Course Outcomes:

After successful completion of the course, the student will have the ability to -

- Apply object-oriented concepts to design solution in C++
- Implement linear data structures (Stack, Queue) in C++
- Implement non-linear data structures (Tree, Graph) in C++

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X		X				X			X

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Evaluation Scheme (Tentative):

- Lab Assignment : 40%
- In-Semester practical Exams (II): 30%
- End-Semester Practical Exam: 30%

Tentative Lecture Distribution:

Week	Lecture	No. of Lectures
Week-1 [5 Jan 2026]	Programming Paradigms, Structure of a C++ program, C++ program compilation, C++ for data structure C++ Data types, Variables, Constants, Expressions, and Namespaces, Control Structure, Array, String.	2
Week-2 [12 Jan 2026]	C++ Function: Pass-by-value, pass-by-reference Default arguments, Inline function, Scope, Recursion, Pointers in c++, Dynamic memory and new and Delete operator C++: Structures and Classes Classes and Objects—Encapsulation, Class structure, Access control, static data and static function	2
Week-3 [19 Jan 2026]	Constructors/Destructors in C++: Multiple constructor, Parametrized constructor, Copy constructor, Destructor	1
Week-4 [26 Jan 2026]	Implementing Linked Lists using C++.	1
Week-5 [2 Feb 2026]	Inheritance: , Types of inheritance, Base/derived classes, Access control & constructor order.	1
Week-6 [9 Feb 2026]	First In-Semester Examination (9 th Feb - Friday to 12 th Feb - Tuesday)	
Week-7 [16 Feb 2026]	Encapsulation and Friend Function	1

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Week-8 [23 Feb 2026]	Virtual Functions & Polymorphism : Run-time binding (late binding), Abstract class Template: Function Template, Class Template. .	1
Week-9 [2 Mar 2026]	In-semester Break (Entire week)	
Week-10 [9 Mar 2026]	Exception Handling: Try, Catch, Throw, Handling run-time errors	1
Week-11 [16 Mar 2026]	Second In-Semester Examination (18 th Mar Wednesday to 23 rd Mar Monday)	
Week-12 [23 Mar 2026]	Implementing Binary Tree using C++.	1
Week-13 [30 Mar 2026]	Introduction to STL: Containers (vector, lists, Stacks, Queues, Deque, Sets, Maps)	1
Week-13 [6 Apr 2026]	Introduction to STL: Iterators and Algorithms, lambda expressions	1
Week-14 [13 Apr 2026]	Implementing Heap Sort using C++	1
Week-15 [20 Apr 2026]	Implementing BFS and DFS using C++	1
Week-16 [27 Apr 2024]	Last date of Classes, Labs, and Tutorials (29 th Apr 2026)	
End-Semester Examination (30 th Apr 2026 to 8 th May 2026)		